

Problems in Rank Correlation

1)

Ten students got the following percentage of marks in Mathematics and Physical sciences:

<i>Students:</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>
<i>Marks in</i>										
<i>Mathematics:</i>	78	36	98	25	75	82	90	62	65	39
<i>Marks in</i>										
<i>Phy. Sciences:</i>	84	51	91	60	68	62	86	58	63	47

Calculate the rank correlation coefficient.

Ans: 0.8424

2)

Ten competitors in a beauty contest were ranked by three judges as follows:

	<i>Competitors</i>									
<i>Judges</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>
<i>A:</i>	6	5	3	10	2	4	9	7	8	1
<i>B:</i>	5	8	4	7	10	2	1	6	9	3
<i>C:</i>	4	9	8	1	2	3	10	5	7	6

Discuss which pair of judges have the nearest approach to common taste of beauty.

3)

Find the rank correlation coefficient between the ranks of the variable X and Y:

<i>X:</i>	10	15	12	17	13	16	24	14	22
<i>Y:</i>	30	42	45	46	33	34	40	35	39

4)

The competitors in a musical contest were ranked by the three judges A, B, C in the following order:

<i>Rank by A:</i>	1	6	5	10	3	2	4	9	7	8
<i>Rank by B:</i>	3	5	8	4	7	10	2	1	6	9
<i>Rank by C:</i>	6	4	9	8	1	2	3	10	5	7

Using rank correlation technique, find which pair of judges have more or less the same taste in music.

5)

The marks secured by recruits in the selection test (X) and in the proficiency test (Y) are given below :

<i>Serial No :</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>
<i>X :</i>	10	15	12	17	13	16	24	14	22
<i>Y :</i>	30	42	45	46	33	34	40	35	39

Calculate the rank correlation co-efficient.